

UHG A4-120
Bielefeld University
Universitätstraße 25
33615 Bielefeld, Germany
☎ +49 521-106-4918

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✉ gerrit.bauch@online.de
🏠 <https://gbauch.github.io/>
🆔 0000-0002-5630-317X

Gerrit Bauch

game theorist, economist, mathematician

👤 professional profile

I am an economist with a strong mathematical background and a passion for game and decision theory. My academic work deals with games and mechanisms that utilize uncertainty for an economic cause and overcome misalignments of theory and observations in practice. The applications of my research span the communication of narratives, language formation, behavioral axiomatization of correlation perception and partnership dissolution mechanisms.

🎓 education

Bielefeld University, Germany 2019–2024

Dr. rer. pol (Ph.D. in economics), passed with distinction (*summa cum laude*).

“From words of approach to words of departure: Essays on noisy communication and partnership dissolution under uncertainty”

Supervisors: Prof. Dr. Riedel, Bielefeld University and Prof. Andreas Blume, PhD, University of Arizona.

Heidelberg University, Germany 2015–2019

M.Sc. in mathematics with economics as field of application. Overall grade: 1,1.

“Über eine Vermutung aus der Theorie Siegelscher Modulformen im Spezialfall $g = 2$ und $k = 5/2$ ”.

Supervised by Prof. Dr. Rainer Weissauer, Heidelberg University.

Heidelberg University, Germany 2012–2015

B.Sc. in mathematics with economics as field of application. Overall grade: 1,4.

“Algebraisch-Geometrische Codes”.

Supervised by Prof. Dr. Kay Wingberg, Heidelberg University.

🏢 research positions & work experience

Bielefeld University, Germany 2024–actual

postdoctoral researcher

Center for Mathematical Economics, Center for Uncertainty Studies (CeUS), RTG Coping with Uncertainty in Dynamic Economies (CUDE), Quantifying and Managing Uncertainty (QUAMU), CRC 1283: Taming Uncertainty.

Bielefeld University, Germany 2019–2024

PhD student

Center for Mathematical Economics and the Center for Uncertainty Studies (CeUS).

Bielefeld University, Germany 2020–2023

IT support at the Center for Mathematical Economics. Tasks included general user assistance, website building (XML based CMS Roxen 5 & 17) and administration, AD & GPO management and hardware and software provisioning (ivanti landesk, citrix workspace).

Heidelberg University, Germany 2014

undergraduate research assistant

Chair of Labor Economics of Prof. Dr. Christina Gathmann, Alfred-Weber institute

📌 scientific output

publications

Gerrit Bauch and Frank Riedel. “The Texas Shoot-Out under Knightian uncertainty”. In: *Games and Economic Behavior* 146 (2024), pp. 35–50. ISSN: 0899-8256. DOI: <https://doi.org/10.1016/j.geb.2024.04.009>. URL: <https://www.sciencedirect.com/science/article/pii/S0899825624000605>.

job market paper

Gerrit Bauch and Manuel Foerster. *Strategic communication of narratives*. Current version available at <https://gbauch.github.io/assets/pdf/narratives.pdf>. 2025. arXiv: 2410.23259 [econ.TH].

R&R

Gerrit Bauch. “Effects of Noise on the Concept Formation of Languages”. In: *Center for Mathematical Economics Working Papers* (2021). R&R (minor revision, 2nd round) in the *International Journal of Game Theory*. Part of PhD thesis. Current version available at https://gbauch.github.io/assets/pdf/Effects_of_noise.pdf.

working papers

Gerrit Bauch and Manuel Foerster. *Strategic communication of narratives*. Current version available at <https://gbauch.github.io/assets/pdf/narratives.pdf>. 2025. arXiv: 2410.23259 [econ.TH].

Gerrit Bauch and Lorenz Hartmann. *Dependence uncertainty: a decision-theoretic approach*. 2025. arXiv: 2503.13416 [econ.TH].

Gerrit Bauch and Lars Müller. “Calling the Price or Making the Call? Role Preferences in Texas Shootout Clauses”. In: (2026). Available at <https://pub.uni-bielefeld.de/record/3013519>.

Gerrit Bauch. “Underreaction and dynamic inconsistency in communication games under noise”. In: *arXiv* (2023). Part of PhD thesis. arXiv: 2311.12496 [econ.TH].

Gerrit Bauch. *Independence and indifferent points imply continuity*. 2025. arXiv: 2411.17883 [econ.TH].

work in progress

Gerrit Bauch, Arthur Dolgoplov, and Manuel Foerster. *Strategic communication of narratives: an experiment*. An experimental study on how individuals strategically implement narratives and how their peers react. Data collection of the experiment running until mid November 2025 at the WiSo lab in Hamburg. 2025.

Gerrit Bauch and Manuel Foerster. *Robust communication of narratives*. A network-based extension of the 2021 AER paper *Using Models to Persuade* by Schwarzstein & Sunderam. 2025.

Gerrit Bauch and Guillaume Pommey. *The economic role of shotgunfunds on efficient partnership dissolutions under liquidity constraints*. A theoretical analysis of how shotgunfunds improve efficiency of partnership dissolutions when co-owners are liquidity constraint. 2025.

Gerrit Bauch, Max Nendel, and Alessandro Sgarabottolo. *Classification of extreme points of probability measures under linear constraints by absolute continuity and f -divergencies*. We provide two characterizations of the extreme points of probability measures under linear constraints for arbitrary measurable spaces: one as a suitable minimum w.r.t. absolute continuity, one by local maximizers of certain f -divergencies. 2025.

dissertation

Gerrit Bauch. “From words of approach to words of departure: Essays on noisy communication and partnership dissolution under uncertainty”. In: *Bielefeld University library* (2024). Publicly available at <https://pub.uni-bielefeld.de/record/2988256>.

research stays

University of Basel, Switzerland

2024 & 2025

Guest researcher at Basel University, March 18–22, 2024 & February 23–March 2, 2025. Invited by Georg Nöldeke and Lorenz Hartmann.

University of Arizona, USA

2022

Research stay from January to May 2022 at the Eller College of Management, University of Arizona. Financial support through a DAAD scholarship and a BGTS mobility grant. Invited by Prof. Andreas Blume, PhD.

referee activities

Economic Theory, Games and Economic Behavior, International Journal of Approximate Reasoning, International Journal of Game Theory, Journal of Economic Behavior & Organization

grants, scholarships, nominations & rewards

- ZiF Focus Group Workshop for QUAMU in March 2026, joint with Frank Riedel & Maren Schmeck (24.000€) **2025**
- Establishing QUAMU as a Bielefeld Focus Area, joint with Frank Riedel & Maren Schmeck (50.000€/year, 3 years) **2025**
- Best Teaching Price (Tutorial Microeconomics 2, graduate) **summer 2025**
- French professorial qualification (qualification MCF) in economics (section 5) and mathematics (section 25) **2025**
- Best Teaching Price (Tutorial Microeconomics 1, graduate) **winter 2024/2025**
- Best Dissertation Prize of Bielefeld University (1.000€) **2024**
- DAAD doctoral scholarship (1.800€/month, travel expenses, health insurance, 4 months) **2022**
- BGTS mobility grant (3.000€) **2022**

teaching

Bielefeld University

- coordinator of the QUAMU Research Seminar **summer 2026**
- coordinator of the QUAMU Research Seminar **winter 2025**
- teaching assistant for Microeconomics 2 (graduate), coordinator of the QUAMU Research Seminar **summer 2025**
- teaching assistant for Microeconomics 1 (graduate), coordinator of the Economic Theory Lunch Seminar **winter 2024**
- *instructor* for an exercise course on Risk and Insurance (undergraduate) including preparing exercise sheets and oral examinations **summer 2023 & 2024**
- *instructor* for a reading group on “Recent developments in game theory and general equilibrium theory” (graduates, PhD candidates, postdocs) **winter 2022**
- *teaching assistant* in mathematics for economists 2 **winter 2021**
- *teaching assistant* in microeconomics 2 (graduate) **summer 2020**
- *tutor* for game theory (undergraduate) **winter 2019**

Heidelberg University

- *teaching assistant* in complex analysis 1 (undergraduate)
including preparing exercise sheets and written examinations **summer 2018**
- *teaching assistant* in complex analysis 2 (undergraduate)
including preparing exercise sheets and oral examinations **winter 2016**
- *tutor* for real analysis, algebra, complex analysis (undergraduate) **2015–2019**

talks & meetings

past talks & meetings

- *Dependence Uncertainty* **2026**
Talk at the The 24th Conference of the International Federation of Operational Research Societies in Linz, Austria, July 13, 2026.
- *Calling the Price or Making the Call? Role Preferences in Texas Shootout Clause* **2026**
Talk at the Conference on Mechanism and Institution Design in York, England, June, 2026.
- *Dependence Uncertainty* **2025**
Talk at the *II CUDE Retreat* in Bergisch Gladbach, September 24, 2025.
- *Strategic communication of narratives* **2025**
Talk at the Annual Conference 2025 of the Germany Economic Association (Verein für Socialpolitik) in Cologne, Germany, September 15, 2025.
- *Quantifying and Managing Uncertainty* **2025**
Pitch introducing the Bielefeld Focus Area Quantifying and Managing Uncertainty (QUAMU) to the administrative staff at Bielefeld University on September 4, 2025.
- *Correlation Uncertainty: a decision-theoretic approach* **2025**
Talk and poster presentation at the 14th International Symposium on Imprecise Probabilities (ISIPTA) in Bielefeld, July 15, 2025.
- *Strategic communication of narratives* **2025**
Talk at the XXIV SAET Conference on Ischia, Italy, July 1, 2025.
- *Strategic communication of narratives* **2025**
Talk at the 14th Lisbon Meetings in Game Theory and Applications, Lisbon, Portugal, June 20, 2025.
- *Correlation Uncertainty: a decision-theoretic approach* **2025**
Talk at the Workshop on “Unforeseen Contingencies: Representations and Applications” in Grenoble, France, June 5, 2025.
- *Strategic communication of narratives* **2025**
Talk at the Seminar of Economics and Management (SEAM) at Paderborn University, Germany, May 18, 2025.
- *Strategic communication of narratives* **2025**
Talk at the Groupe d’Analyse et de Théorie Economique (GATE) at Saint-Étienne University, France, May 13, 2025.
- Guest of the *Standing Committee for Economic Theory* of the Germany Economic Association (Verein für Socialpolitik), May 9-10 in Berlin, Germany. **2025**
- *Correlation Uncertainty* **2025**
Talk at the 2nd Workshop on Learning Under Weakly Structured Information in Tübingen, Germany, April 8, 2025.
- *Strategic communication of narratives* **2025**
Talk at the ADRES conference in Strasbourg, France, January 30, 2025.
- *Strategic communication of narratives* **2024**
Talk at the LEMMA séminaire in Paris Panthéon-Assas, France, October 8, 2024.

- *An academic stay abroad* 2024
Introduction to planning a stay abroad during the PhD at the *I CUDE Retreat* in Willingen, October 1, 2024.
- *The Texas Shoot-Out under Knightian Uncertainty* 2024
Talk at the Conference on Mechanism and Institution Design in Budapest, Hungary, July 12, 2024.
- *The Texas Shoot-Out under Knightian Uncertainty* 2024
Talk at the ESEM 2024 in Rotterdam, Netherlands, August, 2024.
- *Effects of noise on the grammar of languages* 2024
Talk at the 2nd Paris Workshop on Games, Decisions, and Language in Paris, France, on June 13, 2024.
- *Ambiguity aversion and the direction of dynamic inconsistency in sender-receiver games of common interest* 2023
Talk at the SAET 2023 in Paris, France, on July 17, 2023).
- *The Texas Shoot-Out under Knightian Uncertainty* 2023
Talk at the Summer School on Risk and Uncertainty in Economics, Insurance and Finance 2023 in Bielefeld, Germany, on July 7, 2023.
- *The Texas Shoot-Out under Knightian Uncertainty* 2023
Talk at the RUFE 2023 conference in Johannesburg, South Africa, on April 4, 2023.
- *The Texas Shoot-Out under Knightian Uncertainty* 2022
Talk at the Stony Brook International Conference on Game Theory in Stony Brook, Long Island, on July 21, 2022.
- *The Texas Shoot-Out under Knightian Uncertainty* 2022
Online talk at the SAET 2022 at the ANU, Australia, on July 21, 2022.
- *The Texas Shoot Out under Knightian Uncertainty* 2022
Talk at the economics seminar at the Eller College of Management at the University of Arizona, Tuscon on March 23, 2022.
- *What did you say? Introducing a Metric-Dependent Stochastic Noise to Voronoi Languages* 2022
IBE/Brown bag seminar at the Eller College of Management at the University of Arizona, Tuscon on February 10, 2022.
- *Effects of noise on the grammar of Voronoi languages* 2021
Online talk the advanced seminar on general linguistics at the University of Tübingen, Germany, on May 4, 2021.

self-organized

- *ZiF Focus Group Workshop* 2026
Successful grant application for an international ZiF Focus Group for the QUAMU Focus Area with scientific talks, panel discussions, and team-building sessions from March 2-15, 2026 in Bielefeld, Germany. Budget: 24.000€.
- *15th BiGSEM Doctoral Workshop in Economics, Finance and Management* 2020
Self-organized doctoral workshop at the Center for Interdisciplinary Research (ZiF) in Bielefeld, Germany on December 14-15 2020. Budget: 4.000€.

technical skills

sagemath, maple, mathematica, L^AT_EX, python, html, haskell, C++, desktop management with ivanti landesk (citrix), Roxen 17 (CMS)

languages

German (native), English (business competence), Swedish (fluent), French (upper intermediate, B2), Chinese (intermediate, HSK 4)

extracurricular activities

Academics

- Elected representative of the academic mid-level at the Faculty of Economics and Business Administration at Bielefeld University **2024–actual**
- Elected mid-level Faculty representative of the Bielefeld Graduate School of Economics and Management **2024–actual**
- Doctoral Representative of the Bielefeld Graduate School of Economics and Management from May 2021 to October 2023 **2021–2023**

Other

- Member of the Weisser Ring e.V., German non-profit volunteer association for victim assistance and crime prevention **2025–actual**
- Member of the German-Swedish Society, Heidelberg e.V. **2019–actual**

hobbies

 hiking, table tennis, apnea diving, climbing, wave surfing

 learning new languages

 admirer of the poems by Wallace Stevens

references

For reference you may contact:

Prof. Dr. Frank Riedel
✉ frank.riedel@uni-bielefeld.de
Bielefeld University
UHG B4-134/136
Universitätsstraße 25
33615 Bielefeld
Germany

Prof. Andreas Blume, PhD
✉ ablume@arizona.edu
University of Arizona
McClelland Hall 401NN
1130 East Helen Street
Tucson, Arizona 85721-0108
USA

Prof. Dr. Yves Breitmoser
✉ yves.breitmoser@uni-bielefeld.de
Bielefeld University
UHG V8-229
Universitätsstraße 25
33615 Bielefeld
Germany